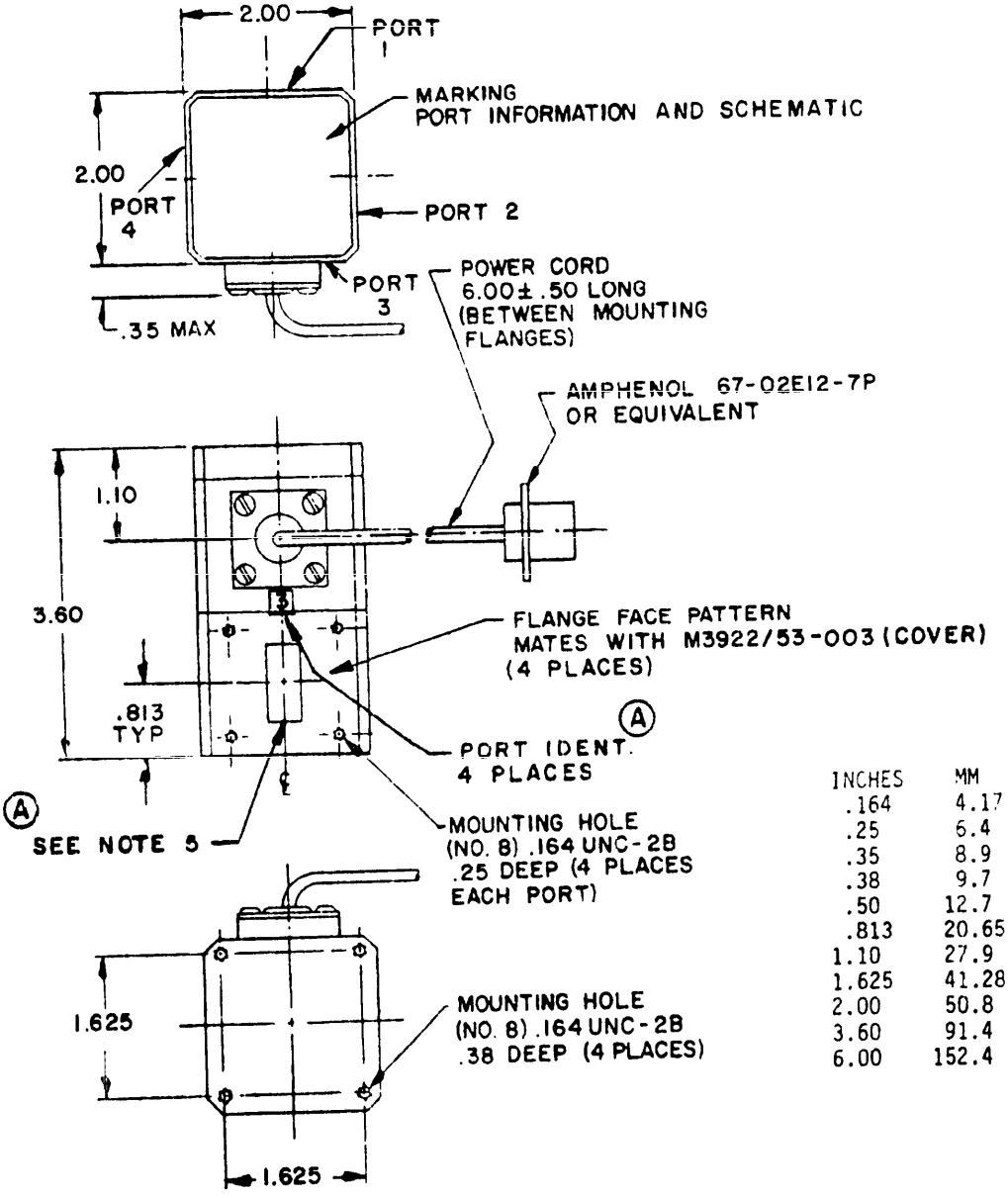
MILITARY SPECIFICATION SHEET

Ⓐ SWITCHES, WAVEGUIDE, 2P2T, FAIL-SAFE, 115 VOLTS, AC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-S-55041.



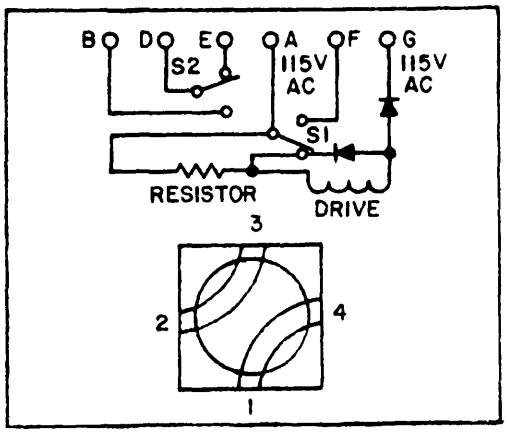
INCHES	MM
.164	4.17
.25	6.4
.35	8.9
.38	9.7
.50	12.7
.813	20.65
1.10	27.9
1.625	41.28
2.00	50.8
3.60	91.4
6.00	152.4

FIGURE 1. Switch configuration and schematic, part no. M55041/8-001.

Ⓐ denotes changes

FSC 5985

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:18Z
Check the source to verify that this is the current version before use.



DEENERGIZED POSITION

IN THE ENERGIZED POSITION PORT 2 CONNECTS TO PORT 1
AND PORT 3 CONNECTS TO PORT 4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-078.
6. DC polarity must be observed.

Ⓐ FIGURE 1. Switch configuration and schematic, part no. M55041/8-001 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:18Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Dash number	Figure number	Housing	Operating frequency range	Nominal operating voltage	Operating current (max)	Holding current (max)	Isolation (min)	VSWR (max)	Transient interference provisions
				ac	amperes	amperes	dB		
001	1	1/ 2/	8.20-12.4	115	.9	.25	50	1.10:1	yes

TABLE I. Electrical performance characteristics - Continued.

Dash number	Insertion loss (max)	Switching time	Life cycles (X1000)	Weight (max)
	dB	second		oz
001	0.1	.30	100	24

1/ Enclosed, sand- and dust-proof when waveguide flanges are capped.

2/ Rotor and housing of aluminum alloy.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Power connectors: See figure 1.

Operating temperature range: 0°C to +55°C.

Electrical and performance characteristics: See table I.

RF power handling capability: Peak - 1 kW with a duty cycle of 1 x 10^{-3}; average - 1 W.

Solenoid switching power: 110 W; holding power - 8.5 W.

Part number: M55041/8-001.

(A) Substitutability data: The switch covered by this specification sheet is substitutable for the user's or manufacturer's part numbers specified in table II. This information in no way implies that the manufacturer's part number or NSN is suitable as a substitute for the military part number.

TABLE II. Substitutability data.

Dash number	Manufacturer's code	Commercial type	NSN 5985
001	82577	253833	---
001	01220	DBGX-563AE-3	---

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:18Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, SM
Air Force - 11, 17, 99
DLA - ES

User activities:
Navy - AS, MC, OS, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-0975-5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:18Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Naval Electronic Systems Command
Washington, DC 20363

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 202640-22T22:18Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-55041/8A	Switches, Waveguide, 2P2T, Fail-saft, 115 volts, AC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:18Z
Check the source to verify that this is the current version before use.